On Algebraic Integral For Dummies

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Abstract

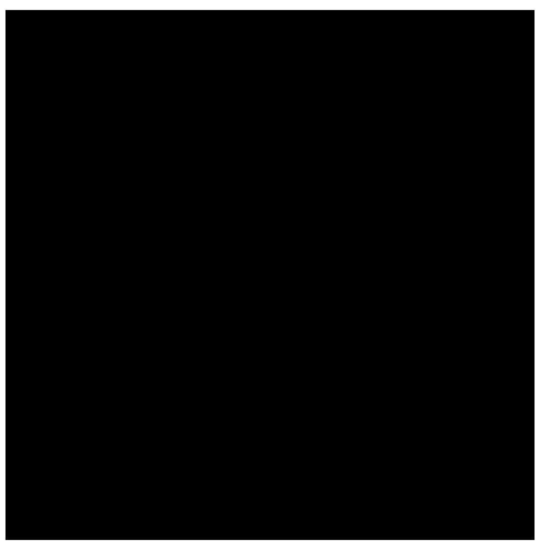
We explain the essence of the integral axiom and provide examples of integrating elementary functions using the new integration method.

Introduction

With his famous painting Black Square, Kazimir Malevich urged artists to embrace simplicity and to avoid multiplying entities beyond necessity — as is well known, Malevich's black square symbolizes economy. Like Malevich, with this paper I want to urge mathematicians toward economy of meaning and the pursuit of simplicity — extra complexity distances them from "ordinary people", while the universe is beautiful in its simplicity, and practice shows that any mathematical concept can be explained to a child if one understands it well enough.

The integration method I propose in this paper is simple and economical. I am writing it in response to requests from several people who told me that the first paper [1] was too complicated and that the practical application of the theory of the algebraic integral was not obvious. And although doing mathematics in today's realities seems to me extremely immoral, I nevertheless decided to answer these requests with this paper, in order to bring the matter to some logical conclusion. In this paper, I will explain in simple words the essence of the theory of the algebraic integral and show how to integrate functions using a new method that is much simpler than the classical one. The substitution method currently used to solve integrals is complicated — its application is an art, mastering which causes considerable difficulties for students. Moreover, using this method assumes that we can replace a function with an independent variable — students usually have a rather vague idea of the nature of this operation, but in reality, justifying the method requires invoking concepts from topology and differential geometry — such as limit, continuity, and smoothness. The new integration method proposed in this paper uses pure algebra, without appealing even to the concept of a limit — either explicitly or implicitly. Furthermore, in the conclusion, I will suggest some other, most obvious applications of the algebraic integral.

Beyond being a symbol of economy, Malevich's Black Square can also be seen as a symbol of a completed theorem or even a theory — depending on the size of the square — as well as a symbol of a tombstone, which in turn symbolizes the completion of a person's life or, for example, of some destructive international organization. I wish to express my support and solidarity with Grigory Perelman — a man who has likely done more for science than anyone else in its entire history, having sacrificed his own well-being for it. I believe this sacrifice was not in vain, and I will do everything in my power to keep it that way.



What is the integral axiom?

In the first part of the paper, we will show that the integral is not a function of a single variable, but an operation $f \triangleright g = \int f\, dg$. While the method of change of variables is nothing other than the distributivity of composition over that operation:

$$(f \triangleright g) \circ h = (f \circ h) \triangleright (g \circ h)$$

Let's begin.

Let $\langle R, \cdot, \circ \rangle$ be a near-ring. That is, a ring with only right distributivity. That is, R such a structure that

$$\forall f, g, h \in R : (f \cdot g) \circ h = (f \circ h) \cdot (g \circ h)$$

Let $d \colon R \to R$ be a function satisfying the chain rule, that is

$$\forall f, g \in R : d(f \circ g) = \big(d(f) \circ g\big) \cdot d(g)$$

Let $\int \colon R \to R$ be a function opposite to d. Let's call this function the **antiderivative**. Then

Theorem. $\forall f, g, h \in R :$

$$\Big(\int f \cdot d(g)\Big) \circ h = \int (f \circ h) \cdot d(g \circ h)$$

Lemma. $\forall f, g \in R :$

$$\int (f) \circ \int (g) = \int \Big(f \circ \int (g) \cdot g\Big).$$

Proof. For any $x, y \in R$, chain rule gives:

$$d(x \circ y) = d(x) \circ y \cdot d(y).$$

Substitute $x = \int f$, $y = \int g$:

$$d\Big(\int f \circ \int g\Big) = \Big(f \circ \int g\Big) \cdot g.$$

Applying $\int$ to both sides yields

$$\int f \circ \int g = \int \big(f \circ \int g \cdot g\big). \quad \square$$

Proof of Theorem:

$$\Big(\int f \cdot d(g)\Big) \circ h = \Big(\int f \cdot d(g)\Big) \circ \int d(h)$$

Applying Lemma:

$$= \int \Big(\big[f \cdot d(g)\big] \circ \Big(\int d(h)\Big) \cdot d(h)\Big)$$

$$= \int \Big(\big[f \cdot d(g)\big] \circ h \cdot d(h)\Big)$$

By right distributivity in R:

$$= \int \Big(\big[f \circ h\big] \cdot \big[d(g) \circ h\big] \cdot d(h)\Big)$$

And reduce the expression by the chain rule for d:

$$= \int \Big(\big[f \circ h\big] \cdot d(g \circ h)\Big)$$

Or in other words:

$$(f \triangleright g) \circ h = (f \circ h) \triangleright (g \circ h)$$

— We call it "integral axiom". ■

There is also a converse statement: informally speaking, for any distributive operation, there exists a corresponding function that satisfies the chain rule. For details see Theorem 2 in "On Algebraic Integral" [1].

Axioms

Before moving on to practical examples of solving integrals using the algebraic method, I will recall the axioms of a **square rring with exponent**, which you can find in Definitions 4.11, 4.10 and 4.4 in [1].

Definition. In particular, the ring of diffeomorphisms $\langle C^1, \cdot, \circ \rangle$ together with the antiderivative as an algebraic integral forms a **square rring with a bilinear integral** $\langle R, +, \cdot, \circ, \triangleright, 0, 1, \mathrm{x}, \mathrm{e} \rangle$, that is $\langle R, \cdot, \circ \rangle$ is a near-ring such that R is vector space over the field C and $\forall f, g, h \in R$, $\forall \alpha, \beta \in C$:

1. $1 \triangleright h = h$ (identity axiom)
2. $(f \cdot g) \triangleright h = f \triangleright (g \triangleright h)$ (associativity)
3. $(f \triangleright h) \circ g = (f \circ g) \triangleright (h \circ g)$ (integral axiom)
4. $(\alpha f + \beta g) \triangleright h = \alpha(f \triangleright h) + \beta(g \triangleright h)$ (linearity in the first argument)
5. $f \triangleright (\alpha g + \beta h) = \alpha(f \triangleright g) + \beta(f \triangleright h)$ (linearity in the second argument)

and the integral $\triangleright$ has an exponential element $\mathrm{e} \in R$ such that $\mathrm{e} \triangleright \mathrm{x} = \mathrm{e}$. In addition we assume that the exponential element is additive:

$$\mathrm{e} \circ (f + g) = (\mathrm{e} \circ f) \cdot (\mathrm{e} \circ g)$$

- to have the Leibniz rule by Theorem 6 from [1].

Remark. It is easy to see that $d(\mathrm{x}) = 1$:

$$d(\mathrm{x}) = d(\mathrm{x} \circ \mathrm{x}) = d(\mathrm{x}) \circ \mathrm{x} \cdot d(\mathrm{x}) = d(\mathrm{x}) \cdot d(\mathrm{x})$$

- therefore we simply write $\int f$ instead of the usual notation $\int f\, dx$. Note that composition has priority over the multiplication.

Remark. It is easy to see that the identity axiom and associativity also hold for the antiderivative. Denoting $d(h) = t$, we get:

1. $\int 1\, dh = \int 1 \cdot t = h$
2. $\int f\, d(\int g\, dh) = \int f\, d(\int g \cdot t) = \int fgt = \int (fg)\, dh$

Examples

We will now integrate several elementary functions using the integral axiom, to demonstrate how the theorem simplifies integration by turning it into a purely algebraic process.

Example 1. $\mathrm{x} \triangleright \ln = \mathrm{x}$.

Denoting by ln the compositional inverse of the exponential element, $\mathrm{e} \circ \ln = \ln \circ \mathrm{e} = \mathrm{x}$, let's solve:

$$\mathrm{x} \triangleright \ln = (\mathrm{e} \circ \ln) \triangleright (\mathrm{x} \circ \ln) = (\mathrm{e} \triangleright \mathrm{x}) \circ \ln$$

- by integral axiom

$$= \mathrm{e} \circ \ln = \mathrm{x} \quad \square$$

Example 2. $\frac{1}{x} \triangleright x = \ln$:

$$\frac{1}{x} \triangleright x = \frac{1}{\mathrm{x}} \triangleright (\mathrm{x} \triangleright \ln)$$

- by Example 1

$$= (\frac{1}{\mathrm{x}} \cdot \mathrm{x}) \triangleright \ln = 1 \triangleright \ln = \ln$$

-by associativity and identity axioms $\square$

Example 3.

$$\frac{1}{\mathrm{x} \cdot \ln} \triangleright \mathrm{x} = \frac{1}{\ln} \triangleright (\frac{1}{\mathrm{x}} \triangleright \mathrm{x}) = \frac{1}{\ln} \triangleright \ln$$

- by associativity axiom and Example 2

$$= (\frac{1}{\mathrm{x}} \circ \ln) \triangleright (\mathrm{x} \circ \ln) = (\frac{1}{\mathrm{x}} \triangleright \mathrm{x}) \circ \ln$$

- apply the integral axiom.

$$= \ln(\ln(\mathrm{x})) \quad \square$$

Example 4.

$$\mathrm{x}^2(\mathrm{x}^3+1)^{\frac{1}{3}} \triangleright \mathrm{x} = (\mathrm{x}^3+1)^{\frac{1}{3}} \triangleright (\mathrm{x}^2 \triangleright \mathrm{x}) = (\mathrm{x}^3+1)^{\frac{1}{3}} \triangleright \frac{1}{3}\mathrm{x}^3$$

- using associativity and integrating x^2

$$= \frac{1}{3}[\mathrm{x}^{\frac{1}{3}} \circ (\mathrm{x}^3+1) \triangleright (\mathrm{x}^3+1)] = \frac{1}{3}[(\mathrm{x}^{\frac{1}{3}} \triangleright \mathrm{x}) \circ (\mathrm{x}^3+1)]$$

- use linearity in the second argument and the axiom of integral. Note that composition has priority over the integral.

$$= \frac{1}{3}[(\frac{3}{4}\mathrm{x}^{\frac{4}{3}}) \circ (\mathrm{x}^3+1)] = \frac{1}{4}(\mathrm{x}^3+1)^{\frac{4}{3}} \quad \square$$

Example 5.

$$\frac{\sin}{1+\cos^2} \triangleright x = \frac{1}{1+\cos^2} \triangleright (-\cos)$$

$$= -(\frac{1}{1+x^2} \circ \cos \triangleright \cos) = -(\frac{1}{1+\mathrm{x}^2} \triangleright \mathrm{x}) \circ \cos$$

- the integral axiom

$$= -\arctan(\cos(\mathrm{x})) \quad \square$$

Remark.
In the case when the exponential element is additive, i.e.

$$\mathrm{e}(f+g) = \mathrm{e}(f) \cdot \mathrm{e}(g)$$

this is equivalent to the Leibniz rule by Theorem 6 in [1]. Given that, as we established above, $d(\mathrm{x}) = 1$, the Leibniz rule uniquely determines d on elements x^n by induction for natural n, with base $n = 2$, and then extends uniquely to rational and real powers. Sine and cosine, in turn, may be defined as elements satisfying the rules: $d(\sin) = \cos$, $d(\cos) = -\sin$, $\sin(0) = 0$, $\cos(0) = 1$ — it is well known that these conditions uniquely determine sine and cosine over the real numbers; the remaining trigonometric functions are defined in terms of these.

Alternatively, in Examples 4 and 5 one may simply regard R as the ring of diffeomorphisms with the standard antiderivative.

Conclusion

Simplicity of finding integrals demonstrated in Examples 1–5 suggests that the proposed algebraic method of integration can be turned into a fast and efficient symbolic integration algorithm for computer algebra systems and symbolic libraries such as MATLAB and SymPy. It remains an open question whether the method can accelerate numerical integration — which, in theory, could lead to the optimization and acceleration of neural networks and other mathematical systems.

References

[1] N. Umerov *On Algebraic Integral.* Zenodo, 2025. DOI: 10.5281/zenodo.17376700